

Dedicated to the Memory of Doctor of Chemical Sciences,
Professor L.M. Yagupol'skii

Synthesis of Polyfluoroaromatic Ketones from Polyfluoroarylzinc Compounds and Acyl Chlorides in the Presence of CuCl

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Abstract—Polyfluoroaromatic ketones Ar_FCOAlk and Ar_FCOPh ($\text{Ar}_F = \text{C}_6\text{F}_5$, 3- and 4- $\text{F}_3\text{CC}_6\text{F}_4$, 4- NCC_6F_4 , 4-EtOCOC $_6\text{F}_4$, $\text{C}_5\text{F}_4\text{N}$, nonafluorindan-5-yl; Alk = Me, Et, *i*-Pr, *t*-Bu) were synthesized by reactions of polyfluoroarylzinc compounds with acyl chlorides in the presence of CuCl.

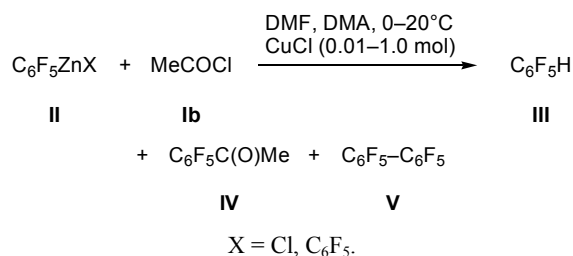
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We previously developed methods for the preparation of polyfluorinated aromatic zinc compounds via reactions of zinc with chloropolyfluoroarenes and perfluoroarenes in dimethylformamide (DMF) [1–4]. Using polyfluoroarylzinc derivatives we obtained polyfluoro- and bromopolyfluoroarenes, perfluorinated biarenes [1–3], and allylpolyfluoroarenes [4]. While studying chemical properties of polyfluorinated aromatic zinc compounds we made an attempt to obtain polyfluoroaromatic ketones via reactions with carboxylic acid chlorides. However, the products of these reactions were *N,N*-dimethylbis(polyfluoroaryl)methanamines [5, 6]. 2,3,4,5,6-Pentafluorobenzophenone was synthesized previously from pentafluoroiodobenzene and benzoyl chloride (**1a**) by the action of Zn/Ag couple deposited on graphite in the presence of $\text{CuCN} \cdot 2\text{LiCl}$ [7]. Polyfluorinated aromatic organocopper compounds obtained from polyfluoroaryllithiums and polyfluoroarylmagnesiums reacted with carboxylic acid bromides, chlorides, and fluorides in diethyl ether or tetrahydrofuran, yielding the corresponding ketones [8, 9]. Kremlev et al. [10] and Rieke et al. [11] recently proposed to synthesize polyfluoroaromatic ketones from pentafluorophenylsilver and tetrafluoropyridylmanganese bromide.

With the goal of developing a general procedure for the preparation of polyfluoroaromatic ketones we examined reactions of polyfluoroarylzinc compounds with acyl chlorides in the presence of copper(I)

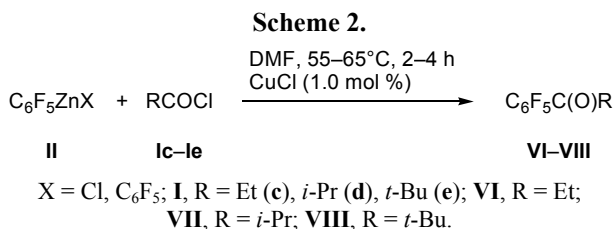
chloride. Polyfluorinated aromatic organozinc compounds were synthesized from chloropolyfluoroarenes and perfluoroarenes and metallic zinc in dimethylformamide in the presence of SnCl_2 [2–4]. The series of chloropolyfluoroarenes and perfluoroarenes included chloropentafluorobenzene, octafluorotoluene, 3-chloroheptafluorotoluene, pentafluorobenzonitrile, ethyl pentafluorobenzoate, pentafluoropyridine, and perfluoroindan. Solutions of organozinc reagents prepared from perfluoroarenes were preliminarily treated with CaCl_2 with a view to convert zinc fluoride into zinc chloride and thus minimize formation of acyl fluorides from acyl chlorides by the action of, e.g., ZnF_2 (it is known that benzoyl fluoride is formed from benzoyl chloride and ZnF_2 [12]). In the reactions of pentafluorophenylzinc chloride and bis(pentafluorophenyl)zinc (**II**) with 1.0 mol of acetyl chloride (**1b**) at $\sim 0^\circ\text{C}$ in the presence of 0.01 mol of CuCl, the major product was pentafluorobenzene (**III**), while pentafluoroacetophenone (**IV**) and decafluorobiphenyl (**V**)

Scheme 1.

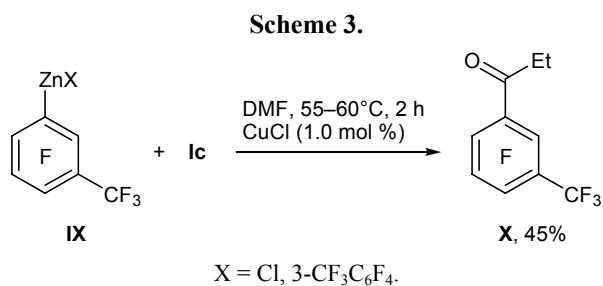


were formed as minor ones. When the amount of CuCl was raised to 0.5–1.0 mol (1.5 mol of **Ib**), ketone **IV** became the major product at 20°C (yield 44–47%). The yield of **IV** increased to 62% when the reaction was carried out in *N,N*-dimethylacetamide (DMA) (Scheme 1).

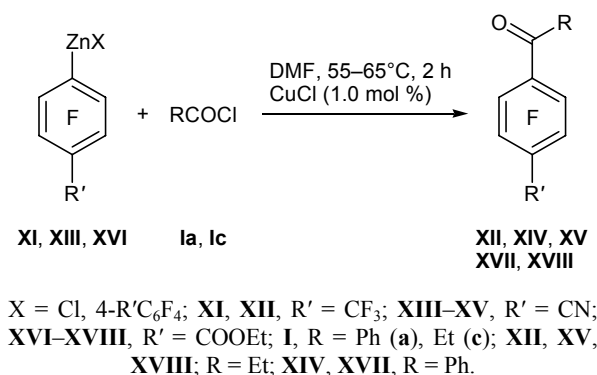
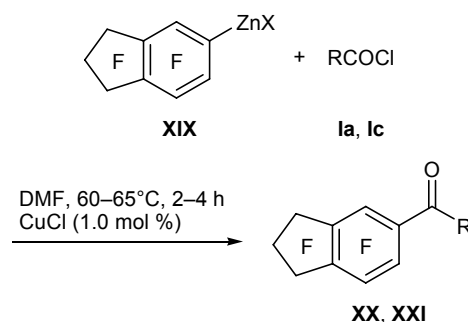
Small amounts of CuCl (~0.01 mol) were sufficient to obtain polyfluoroaromatic ketones as the major products in the reactions of polyfluoroarylzinc compounds with other acyl chlorides. For example, 1-(2,3,4,5,6-pentafluorophenyl)propan-1-one (**VI**) was synthesized in 66% yield by reaction of organozinc compound **II** with propionyl chloride (**Ic**). Analogous ketones **VII** and **VIII** were obtained from compound **II** and isobutyryl and pivaloyl chlorides **Id** and **Ie** in 65 and 71% yield, respectively (Scheme 2). Compounds **III** and **V** were also formed as by-products.



The reaction of propionyl chloride (**Ic**) with 2,4,5,6-tetrafluoro-3-trifluoromethylphenylzinc compounds **IX** in the presence of CuCl gave 1-(2,4,5,6-tetrafluoro-3-trifluoromethylphenyl)propan-1-one (**X**) (Scheme 3).



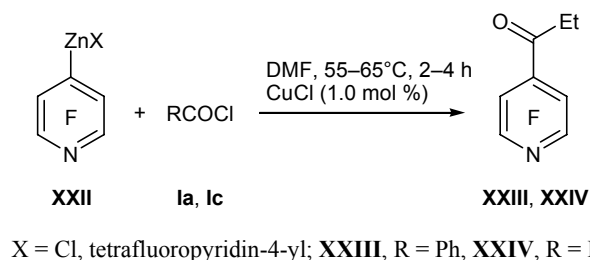
Likewise, propionyl chloride (**Ic**) reacted with 2,3,5,6-tetrafluoro-4-trifluoromethylphenylzinc derivatives **XI** in the presence of CuCl to produce 1-(2,3,5,6-tetrafluoro-4-trifluoromethylphenyl)propan-1-one (**XII**) (Scheme 4). Analogous reactions of polyfluorinated aromatic zinc compounds **XIII** and **XVI** with acid chlorides **Ia** and **Ic** afforded 4-benzoyl-2,3,5,6-tetrafluorobenzonitrile (**XIV**), 4-propionyl-2,3,5,6-tetrafluorobenzonitrile (**XV**), ethyl 4-benzoyl-2,3,5,6-tetrafluorobenzoate (**XVII**), and ethyl 4-propionyl-2,3,5,6-tetrafluorobenzoate (**XVIII**) (Scheme 4).

Scheme 4.**Scheme 5.**

X = Cl, perfluoroindan-5-yl; **XX**, R = Ph, **XXI**, R = Et.

Organozinc compounds **XIX** derived from decafluoroindan reacted with benzoyl chloride (**Ia**) and propionyl chloride (**Ic**) under analogous conditions, and the products were (1,1,2,2,3,3,4,6,7-nonafluoroindan-5-yl)phenylmethanone (**XX**) and 1-(1,1,2,2,3,3,4,6,7-nonafluoroindan-5-yl)propan-1-one (**XXI**), respectively (Scheme 5). Likewise, 2,3,5,6-tetrafluoropyridin-4-yl(phenyl)methanone (**XXIII**) and 1-(2,3,5,6-tetrafluoropyridin-4-yl)propan-1-one (**XXIV**) were synthesized from tetrafluoropyridin-4-ylzinc compounds **XXII** and acid chlorides **Ia** and **Ic** (Scheme 6).

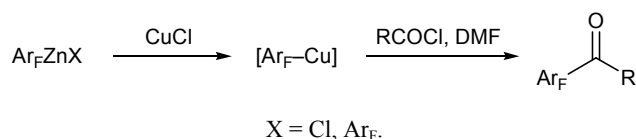
When compound **XXII** was not preliminarily treated with calcium chloride, the amount of propionyl chloride should be increased to 3 equiv to obtain

Scheme 6.

ketone **XXIV** in 62% yield; the reaction of **XXII** with 2 equiv of **Ic** gave only 27% of **XXIV**. Here, 2,3,5,6-tetrafluoropyridine (**XXV**), octafluoro-4,4'-bipyridyl (**XXVI**), and *N,N*-dimethyl-1,1-bis(2,3,5,6-tetrafluoropyridin-4-yl)methanamine (**XXVII**) were formed as by-products. Preliminary treatment of **XXII** with calcium chloride ensured 68% yield of ketone **XXIV** in the reaction with ~1 mol of acid chloride **Ic**. Presumably, treatment of **XXII** with CaCl_2 hampers transformation of acid chloride **Ic** into the corresponding fluoride by the action of ZnF_2 . The formation of propionyl fluoride (**If**) in the reaction mixtures obtained from organozinc compound **XXII** without preliminary treatment with CaCl_2 was confirmed by ^{19}F NMR spectroscopy. Presumably, fluoride **If** is less reactive than chloride **Ic** toward organozinc derivative **XXII**. Weaker reactivity of acetyl and benzoyl fluorides compared to the corresponding chlorides was observed previously in their hydrolysis [13].

Presumably, addition of CuCl changes the reaction direction toward formation of polyfluoroaromatic ketones rather than *N,N*-dimethylbis(polyfluoroaryl)-methanamines [5, 6] (that are formed from polyfluoroarylzinc compounds and acyl chlorides in DMF with participation of Vilsmeier reagent) due to transmetalation process with intermediate generation of polyfluoroarylcopper compounds (Scheme 7).

Scheme 7.



According to the ^{19}F NMR data, treatment of **II** with CuCl gave assumingly pentafluorophenylcopper. No Vilsmeier reagent is likely to be formed in the reactions of polyfluoroarylzinc compounds with CuCl and acyl chlorides, leading to polyfluoroaromatic ketones. In the reaction of $2\text{-NCC}_6\text{H}_4\text{Cu(CN)ZnI}$ (prepared from the corresponding organozinc reagent and $\text{CuCN} \cdot 2\text{LiCl}$) with benzoyl chloride in DMF, no ketone was obtained [14], though Vilsmeier reagent and product of its reaction with the organocopper compound were formed.

The formation of polyfluoroarenes like **III** [6] and compound **XXVII** [5, 6] may be rationalized in a similar way. The different behaviors of reagent **II** in the reactions with acyl chlorides **Ib** and **Ic** in the presence of 0.01 mol of CuCl in DMF (in the first case, com-

pound **III** was the major product, and in the second, ketone **VI**) may be due to higher reactivity of **II** as a base toward chloride **Ib** rather than **Ic** (the C–H bond in **Ib** is more acidic than in **Ic**; cf., e.g., the data for CH acidities of acetone and diethyl ketone [15]). This assumption does not contradict the results of the reaction of **II** with acetyl chloride **Ib** in DMA in the absence of CuCl , which gave compound **III**.

Thus we have developed a general procedure for the synthesis of polyfluoroaromatic ketones by reactions of polyfluoroarylzinc compounds with acyl chlorides in the presence of CuCl .

EXPERIMENTAL

The NMR spectra were recorded on a Bruker AC-300 spectrometer at 282.4 MHz for ^{19}F and 300 MHz for ^1H from solutions in dimethylformamide (reaction mixtures containing organozinc compounds) or carbon tetrachloride (in other cases). The chemical shifts were determined using hexafluorobenzene (δ_F –162.9 ppm relative to CCl_3F) and hexamethyldisiloxane (δ 0.04 ppm relative to tetramethylsilane) as internal references. The IR spectra were measured from solutions in carbon tetrachloride on a Bruker Vector 22 spectrometer. The UV spectra were recorded on a Hewlett–Packard 8453 spectrophotometer from solutions in ethanol. The molecular weights and elemental compositions of the products were determined by high-resolution mass spectrometry on Finnigan MAT 8200 and DFS instruments (electron impact, 70 eV). The GC–MS data were obtained on a Hewlett–Packard G1800A system consisting of an HP 5890 Series II gas chromatograph and an HP 5971 mass selective detector (HP-5 capillary column, 30 m $\times$ 0.25 mm, stationary phase 5% of diphenyl- and 95% of dimethylpolysiloxane; carrier gas helium, flow rate 1 ml/min; injector temperature 280°C; ion source temperature 173°C, electron impact, 70 eV, a.m.u. range 30–650). Gas-liquid chromatography was performed on a Hewlett–Packard 5980 instrument equipped with an HP-5 quartz capillary column, 30 m $\times$ 0.52 mm (stationary phase dimethyldiphenylpolysiloxane, film thickness 2.6 μm) and a thermal conductivity detector. The melting points were determined on a Kofler hot stage.

Organozinc compounds **II**, **IX**, **XI**, **XIII**, **XVI**, **XIX**, and **XXII** were synthesized according to the procedures reported in [4]. Solutions of compounds **XI**, **XIII**, **XVI**, **XIX**, and **XXII** were treated with an equimolar amount of calcium chloride (preliminarily calcined for 3 h at 250–300°C), the mixture was stirred

for 3 h at room temperature, the solution was separated by decanting, and the precipitate was washed with DMF. The resulting solutions were weighed, and the concentration of organozinc reagent was calculated with correction for its relative concentration on the basis of the ^{19}F NMR data. These solutions contained $\text{Ar}_\text{F}\text{ZnCl}$, $(\text{Ar}_\text{F})_2\text{Zn}$, and $\text{Ar}_\text{F}\text{H}$ as impurity. A solution containing 91% of compounds **II** and 9% of **III** was used in the synthesis of **IV**, **VI**, and **VII**,

Organozinc compounds **II** in *N,N*-dimethylacetamide were prepared as described in [4].

1-(2,3,4,5,6-Pentafluorophenyl)ethanone (IV).

a. A solution of organozinc compounds **II**, 7.14 g (10.0 mmol; hereinafter, the amount of reagent per aromatic ring is given), was added under argon to 0.01 g (0.10 mmol) of CuCl . The mixture was cooled with an ice bath, 0.79 g (10.0 mmol) of acetyl chloride (**Ib**) was added dropwise under stirring over a period of 5 min, and the mixture was stirred for 30 min on cooling with an ice bath. Water, 10 ml, was added, and the mixture was subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.01 g of a mixture containing (according to the GLC data) 12% of ketone **IV** (yield 6%), 82% of pentafluorobenzene (**III**), and 2% of decafluorobiphenyl (**V**).

b. A solution of organozinc compounds **II**, 7.14 g (10.0 mmol), was added under argon to 0.5 g (5.0 mmol) of CuCl , and the mixture was stirred for 1 h. According to the ^{19}F NMR data, the mixture contained 57% of assumed pentafluorophenylcopper compound (δ_F 52.5 ppm, m [16]), 31% of organozinc compounds **II**, 9% of **III**, and 3% of **V**. Acetyl chloride (**Ib**), 1.18 g (15.0 mmol), was then added dropwise over a period of 5 min under continuous stirring at room temperature, and the mixture was stirred for 30 min, treated with 10 ml of water, and subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.33 g of a mixture containing (according to the GLC data) 70% of ketone **IV** (yield 44%), 21% of **III**, and 5% of **V**. The mixture was distilled under reduced pressure (~ 3 mm), a fraction with bp 39.5–40°C being collected (0.6 g); it contained (GLC) $\sim 99\%$ of ketone **IV** (yield 29%). IR spectrum, ν , cm^{-1} : 1716, 1649 ($\text{C}=\text{O}$) [17]. ^1H NMR spectrum, δ , ppm: 2.56 t (3H, CH_3 , $J = 1.8$ Hz). ^{19}F NMR spectrum, δ_F , ppm: 1.4 m (2F, 3-F, 5-F), 11.7 t.t (1F, 4-F, $^3J_\text{FF} = 20.4$, $^4J_\text{FF} = 3.5$ Hz), 20.6 m (2F, 2-F, 6-F) [18].

c. The reaction was carried out as described above in *b* using 1.0 g (10.1 mmol) of CuCl . Before addition

of acetyl chloride (**Ib**), the reaction mixture contained (^{19}F NMR) 85% of assumed pentafluorophenylcopper compound, 4% of organozinc compounds **II**, 8% of **III**, and 3% of **V**. After appropriate treatment, we isolated 1.32 g of a mixture containing (GLC) 74% of ketone **IV** (yield 47%), 17% of **III**, and 5% of **V**.

d. A mixture of 4.63 g (5.0 mmol) of organozinc compounds **II** in DMA (relative concentration of **II** 91% with an impurity of 9% of **III**) and 1.57 g (10.0 mmol) of acetyl chloride (**Ib**) was kept for 1 h at 20°C. According to the ^{19}F NMR data, the mixture contained only compound **III**.

e. A mixture of 9.26 g (10.0 mmol) of organozinc compounds **II** in DMA (with the same composition as in *d*), 1.18 g (15.0 mmol) of acetyl chloride (**Ib**), and 0.5 g (5.0 mmol) of CuCl was kept for 1 h at 20°C. According to the ^{19}F NMR data, the mixture contained $\sim 69\%$ of ketone **IV**, $\sim 29\%$ of **III**, and $\sim 2\%$ of **V**. The mixture was treated with 10 ml of water and subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.65 g of a mixture containing (GLC) 79% of ketone **IV** (yield 62%), 17% of **III**, and 2% of **V**.

1-(2,3,4,5,6-Pentafluorophenyl)propan-1-one

(VI). A mixture of 7.14 g (10.0 mmol) of organozinc compounds **II** (a solution in DMF), 1.02 g (11.0 mmol) of propionyl chloride (**Ic**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h under argon at 55–60°C. The mixture was cooled to room temperature, treated with 10 ml of water, and subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.81 g of a mixture containing (GLC) 82% of ketone **VI** (yield 66%) and 14% of **III**. The mixture was distilled under reduced pressure (~ 3 mm), and a fraction with bp 39–42.5°C (0.96 g) was collected; according to the GLC data, it contained $\sim 99\%$ of ketone **VI** (yield 43%). IR spectrum, ν , cm^{-1} : 1721, 1649 ($\text{C}=\text{O}$). ^1H NMR spectrum, δ , ppm: 1.15 t (3H, CH_2CH_3 , $J_\text{HH} = 7.3$ Hz), 2.79 q (2H, CH_2CH_3 , $J_\text{HH} = 7.3$ Hz). ^{19}F NMR spectrum, δ_F , ppm: 1.5 m (2F, 3-F, 5-F), 11.0 t.t (1F, 4-F, $^3J_\text{FF} = 20.5$, $^4J_\text{FF} = 2.5$ Hz), 20.1 m (2F, 2-F, 6-F) [19].

2-Methyl-1-(2,3,4,5,6-pentafluorophenyl)propan-

1-one (VII). A mixture of 9.55 g (13.37 mmol) of organozinc compounds **II** (a solution in DMF), 1.59 g (15.0 mmol) of chloride **Id**, and 0.015 g (0.15 mmol) of CuCl was heated for 4 h at 60–65°C under argon. According to the ^{19}F NMR data, the mixture contained $\sim 88\%$ of ketone **VII**, $\sim 10\%$ of **III**, and $\sim 2\%$ of **V**. The

mixture was cooled, treated with 10 ml of water, and subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 2.47 g of a mixture containing (GLC) 84% of ketone **VII** (yield 65%) and 3% of **III**. The mixture was distilled under reduced pressure (~ 3 mm), and a fraction with bp 68–70°C (1.24 g) was collected; according to the GLC data, it contained $\sim 99\%$ of ketone **VII** (yield 40%). IR spectrum, ν , cm^{-1} : 1720, 1651 ($\text{C}=\text{O}$). ^1H NMR spectrum, δ , ppm: 1.18 d (6H, CH_3 , $J_{\text{HH}} = 7.0$ Hz), 2.97 m (1H, CH). ^{19}F NMR spectrum, δ_{F} , ppm: 1.6 m (2F, 3-F, 5-F), 10.5 t.t (1F, 4-F, $^3J_{\text{FF}} = 20.0$, $^4J_{\text{FF}} = 2.5$ Hz), 20.1 m (2F, 2-F, 6-F) [19].

2,2-Dimethyl-1-(2,3,4,5,6-pentafluorophenyl)propan-1-one (VIII). A mixture of 7.87 g (8.5 mmol) of organozinc compounds **II** (a solution in DMF containing 96% of **II** and 4% of **III**), 1.21 g (10.0 mmol) of 2,2-dimethylpropanoyl chloride (**Ic**), and 0.01 g (0.10 mmol) of CuCl was heated for 4 h at 60–65°C under argon. According to the ^{19}F NMR data, the mixture contained $\sim 75\%$ of ketone **VIII**, $\sim 14\%$ of **III**, and $\sim 11\%$ of **V**. The mixture was cooled to room temperature, 10 ml of water was added, and the mixture was subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 2.02 g of a mixture containing (GLC) 76% of ketone **VIII** (yield 71%), 5% of **III**, and 12% of **V**. The mixture was distilled twice under reduced pressure (~ 3 mm), and a fraction with bp 45–48°C (0.48 g) was collected; it contained (GLC) 94% of ketone **VIII** (yield 19%). The fraction was cooled to $\sim 5^\circ\text{C}$, and crystals separated and were filtered off. We thus isolated 0.11 g of ketone **VIII** with a purity of $\sim 99\%$ (according to the GC–MS data), mp 27–28°C. IR spectrum, ν , cm^{-1} : 2975, 2938, 2909, 2874, 1715, 1653, 1520, 1498, 1479, 1314, 1152, 984, 848. UV spectrum, λ_{max} , nm ($\log \epsilon$): 202 (3.81), 262 (2.84). ^1H NMR spectrum: δ 1.22 ppm, t (9H, CH_3 , $J = 1.1$ Hz). ^{19}F NMR spectrum, δ_{F} , ppm: 1.8 m (2F, 3-F, 5-F), 8.9 t (1F, 4-F, $^3J_{\text{FF}} = 20.5$ Hz), 22.2 m (2F, 2-F, 6-F). GC–MS: m/z 252 $[M]^+$ ($\sim 99\%$). Mass spectrum, m/z (I_{rel} , %): 252 (1) $[M]^+$, 195 (13) $[M - \text{CMe}_3]^+$, 167 (10) $[M - \text{COCMe}_3]^+$, 57 (100) $[\text{CMe}_3]^+$, 41 (37). Found, %: C 52.87; H 3.66; F 37.95. M^+ 252.0582. $\text{C}_{11}\text{H}_9\text{F}_5\text{O}$. Calculated, %: C 52.39; H 3.60; F 37.67. M 252.0568.

1-(2,4,5,6-Tetrafluoro-3-trifluoromethylphenyl)propan-1-one (X). A mixture of 7.78 g (8.48 mmol) of organozinc compounds **IX** (a solution in DMF containing 86% of **IX** and 14% of 3*H*-heptafluorotoluene

XXVIII), 1.02 g (11.0 mmol) of propionyl chloride **Ic**, and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 55–60°C under argon. The mixture was cooled to room temperature, 10 ml of water was added, and the mixture was subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.69 g of a mixture containing (GLC) 63% of ketone **X** (yield 45%) and 22% of **XXVIII**. The mixture was distilled under reduced pressure (~ 2 mm), a fraction with bp 42–44°C (0.73 g) being collected; it contained (GLC) $\sim 96\%$ of ketone **X** (yield 30%). IR spectrum, ν , cm^{-1} : 2987, 2944, 1725, 1645, 1624, 1499, 1332, 1220, 1185, 1157, 902. UV spectrum: λ_{max} 270 nm ($\log \epsilon$ 3.15). ^1H NMR spectrum, δ , ppm: 1.21 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 2.83 q (2H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz). ^{19}F NMR spectrum, δ_{F} , ppm: 1.2 d.d.d (1F, 5-F), 32.5 q.d.d.d (1F, 6-F), 33.7 d.d (1F, 4-F), 44.3 m (1F, 2-F), 105.4 t (3F, CF_3); $J(\text{CF}_3\text{--}2\text{-F}) = 22.6$, $J(\text{CF}_3\text{--}6\text{-F}) = 22.6$, $J(5\text{-F--}6\text{-F}) = 22.2$, $J(4\text{-F--}5\text{-F}) = 22.2$, $J(2\text{-F--}5\text{-F}) = 11.8$, $J(4\text{-F--}6\text{-F}) = 12.2$ Hz. Mass spectrum, m/z (I_{rel} , %): 275 (2), 274 (16) $[M]^+$, 245 (100) $[M - \text{Et}]^+$, 217 (22) $[M - \text{COEt}]^+$, 57 (15) $[\text{COEt}]^+$, 29 (18) $[\text{Et}]^+$. Found, %: C 44.31; H 1.83; F 48.73. M^+ 274.0223. $\text{C}_{10}\text{H}_5\text{F}_7\text{O}$. Calculated, %: C 43.81; H 1.84; F 48.51. M 274.0229.

1-(2,3,5,6-Tetrafluoro-4-trifluoromethylphenyl)propan-1-one (XII). A mixture of 7.35 g (10 mmol) of organozinc compounds **XI** (a solution in DMF containing 97% of **XI** and 3% of 4*H*-heptafluorotoluene **XXIX**), 1.02 g (11 mmol) of propionyl chloride (**Ic**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 60–65°C under argon. The mixture was cooled to room temperature, treated with 10 ml of water, and subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.99 g of a mixture containing (GLC) 93% of ketone **XII** (yield 67%) and 4% of **XXIX**. The product was distilled under reduced pressure (~ 3 mm), and a fraction with bp 49–51°C (1.05 g) was collected; it contained (GLC) $\sim 98\%$ of ketone **XII** (yield 38%). IR spectrum, ν , cm^{-1} : 2987, 2943, 1725, 1605, 1492, 1333, 1193, 1161, 1111, 985. UV spectrum, λ_{max} , nm ($\log \epsilon$): 208 (3.90), 281 (3.38). ^1H NMR spectrum, δ , ppm: 1.23 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 2.84 q (2H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz). ^{19}F NMR spectrum, δ_{F} , ppm: 21.2 m (2F), 23.4 m (2F), 104.9 t (3F, CF_3 , $^4J_{\text{FF}} = 21.8$ Hz). Mass spectrum, m/z (I_{rel} , %): 275 (2), 274 (15) $[M]^+$, 245 (100) $[M - \text{Et}]^+$, 217 (25) $[M - \text{COEt}]^+$, 57 (8) $[\text{COEt}]^+$. Found, %: C 43.62; H 1.75; F 48.80. M^+ 274.0221. $\text{C}_{10}\text{H}_5\text{F}_7\text{O}$. Calculated, %: C 43.81; H 1.84; F 48.51. M 274.0229.

4-Benzoyl-2,3,5,6-tetrafluorobenzonitrile (XIV).

A mixture of 8.93 g (10.0 mmol) of organozinc compounds **XIII** [a solution in DMF containing 91% of **XIII** and 9% of 2,3,5,6-tetrafluorobenzonitrile (**XXX**)], 2.11 g (15.0 mmol) of benzoyl chloride (**Ia**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 55–60°C under argon. According to the ^{19}F NMR data, the mixture contained ~88% of ketone **XIV** and ~12% of **XXX**. The mixture was cooled to room temperature, 10 ml of water was added, and the precipitate was extracted into carbon tetrachloride (3 × 10 ml). The combined extracts were washed with 50 ml of a 10% solution of potassium carbonate and 50 ml of water and dried over MgSO_4 . The solvent was distilled off, and the residue was evacuated. The product was 2.26 g of a solid substance containing (GLC) 92% of ketone **XIV** (yield 75%). Recrystallization from hexane gave 1.79 g (64%) of ketone **XIV** with a purity of ~99% (GLC), mp 81–82°C. IR spectrum, ν , cm^{-1} : 3069, 2247, 1691, 1598, 1491, 1451, 1323, 1236, 993, 812. UV spectrum, λ_{max} , nm ($\log \epsilon$): 200 (4.59), 260 (4.26). ^1H NMR spectrum, δ , ppm: 7.52 m (2H, 3'-H, 5'-H), 7.67 m (1H, 4'-H), 7.79 m (2H, 2'-H, 6'-H). ^{19}F NMR spectrum, δ_{F} , ppm: 23.8 m (2F, 3-F, 5-F), 31.0 m (2F, 2-F, 6-F). Mass spectrum, m/z (I_{rel} , %): 280 (5), 279 (40) $[M]^+$, 202 (4) $[M - \text{Ph}]^+$, 174 (5) $[M - \text{COPh}]^+$, 105 (100) $[\text{COPh}]^+$, 77 (54) $[\text{Ph}]^+$, 60 (17), 51 (25), 43 (25). Found, %: C 60.20; H 1.80; F 26.86; N 4.86. M^+ 279.0303. $\text{C}_{14}\text{H}_5\text{F}_4\text{NO}$. Calculated, %: C 60.23; H 1.81; F 27.22; N 5.02. M 279.0307.

2,3,5,6-Tetrafluoro-4-(1-oxopropyl)benzonitrile (XV). A mixture of 8.93 g (10.0 mmol) of organozinc compounds **XIII** [a solution in DMF containing 91% of **XIII** and 9% of 2,3,5,6-tetrafluorobenzonitrile (**XXX**)], 1.02 g (11.0 mmol) of propionyl chloride (**Ic**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 55–60°C under argon. The mixture was treated with 10 ml of water and steam-distilled, and the organic phase was separated and dried over MgSO_4 to isolate 1.62 g of a mixture containing (GLC) 61% of ketone **XV** (yield 43%) and 35% of **XXX**. Sublimation of the mixture under reduced pressure (128°C, ~9 mm) gave 0.47 g of ketone **XV** (yield 20%) with a purity of ~98% (GLC), mp 40–42°C. IR spectrum, ν , cm^{-1} : 2988, 2944, 2248, 1728, 1489, 1348, 1309, 1134, 985. UV spectrum, λ_{max} , nm ($\log \epsilon$): 201 (4.05), 235 (4.13), 294 (3.36). ^1H NMR spectrum, δ , ppm: 1.24 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 2.83 q (2H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz). ^{19}F NMR spectrum, δ_{F} , ppm: 21.9 m (2F, 3-F, 5-F), 31.0 m (2F, 2-F, 6-F). Mass spectrum, m/z (I_{rel} ,

%): 232 (3), 231 (27) $[M]^+$, 202 (100) $[M - \text{Et}]^+$, 174 (27) $[M - \text{COEt}]^+$, 124 (21), 57 (17) $[\text{COEt}]^+$, 29 (19) $[\text{Et}]^+$. Found, %: C 51.59; H 2.12; F 32.83; N 5.89. M^+ 231.0309. $\text{C}_{10}\text{H}_5\text{F}_4\text{NO}$. Calculated, %: C 51.96; H 2.18; F 32.88; N 6.06. M 231.0307.

Ethyl 4-benzoyl-2,3,5,6-tetrafluorobenzoate (XVII). A mixture of 9.01 g (10.0 mmol) of organozinc compounds **XVI** [a solution in DMF containing 95% of **XVI** and 5% of ethyl 2,3,5,6-tetrafluorobenzoate (**XXXI**)], 1.55 g (11.0 mmol) of benzoyl chloride (**Ia**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 60–65°C under argon. According to the ^{19}F NMR data, the mixture contained ~86% of ketone **XVII** and ~13% of **XXXI**. The mixture was cooled to room temperature, 150 ml of water was added, and the precipitate was filtered off, washed with 50 ml of a 10% solution of sodium carbonate and 50 ml of water, and dried in air. According to the ^{19}F NMR data, the product, 2.15 g, contained ~97% of ketone **XVII** (yield 64%). A 1.00-g sample of the product was subjected to vacuum sublimation (105°C, ~4 mm), followed by recrystallization from hexane, to obtain 0.78 g of **XVII** with a purity of ~99% (GLC), mp 71–71.5°C. IR spectrum, ν , cm^{-1} : 2985, 1746, 1688, 1478, 1322, 1314, 1207, 1053, 993. UV spectrum, λ_{max} , nm ($\log \epsilon$): 203 (4.41), 258 (4.23). ^1H NMR spectrum, δ , ppm: 1.43 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 4.45 q (2H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 7.49 m (2H, 3-H, 5-H), 7.63 m (1H, 4-H), 7.82 m (2H, 2-H, 6-H). ^{19}F NMR spectrum, δ_{F} , ppm: 21.7 m (2F), 23.3 m (2F). Mass spectrum, m/z (I_{rel} , %): 327 (4), 326 (24) $[M]^+$, 281 (15) $[M - \text{C}_2\text{H}_5\text{O}]^+$, 249 (2) $[M - \text{Ph}]^+$, 221 (2) $[M - \text{COPh}]^+$, 105 (100) $[\text{PhCO}]^+$, 77 (25) $[\text{Ph}]^+$, 51 (5). Found, %: C 58.83; H 3.11; F 23.24. M^+ 326.0562. $\text{C}_{16}\text{H}_{10}\text{F}_4\text{O}_3$. Calculated, %: C 58.90; H 3.09; F 23.29. M 326.0561.

Ethyl 2,3,5,6-tetrafluoro-4-(1-oxopropyl)benzoate (XVIII). A mixture of 4.93 g (10.0 mmol) of organozinc compounds **XVI** (a solution in DMF containing 90% of **XVI** and 10% of **XXXI**), 1.02 g (11.0 mmol) of propionyl chloride (**Ic**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 60–65°C under argon. According to the ^{19}F NMR data, the mixture contained ~84% of ketone **XVIII** and ~15% of **XXXI**. It was cooled to room temperature, treated with 10 ml of water, and steam-distilled. The organic phase was separated and dried over MgSO_4 . We isolated 1.42 g of a mixture containing (GLC) 85% of ketone **XVIII** (yield 43%) and 14% of **XXXI**. The product was distilled under reduced pressure (~3 mm), and

a fraction with bp 107–108°C (0.68 g) was collected; it contained (GLC) ~99% of ketone **XVIII** (yield 24%). IR spectrum, ν , cm^{-1} : 2986, 2943, 2908, 1746, 1724, 1477, 1313, 1227, 1117, 985. UV spectrum, λ_{max} , nm ($\log \epsilon$): 204 (3.90), 228 (3.88), 285 (3.14). ^1H NMR spectrum, δ , ppm: 1.21 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 1.40 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.1$ Hz), 2.83 q (2H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 4.41 q (2H, OCH_2 , $J_{\text{HH}} = 7.1$ Hz). ^{19}F NMR spectrum, δ_{F} , ppm: 20.0 m (2F), 23.3 m (2F). Mass spectrum, m/z (I_{rel} , %): 279 (3), 278 (24) $[M]^+$, 249 (100) $[M - \text{Et}]^+$, 233 (15) $[M - \text{C}_2\text{H}_5\text{O}]^+$, 221 (34) $[M - \text{COEt}]^+$, 57 (2) $[\text{COEt}]^+$. Found, %: C 52.26; H 3.53; F 27.50. M^+ 278.0563. $\text{C}_{12}\text{H}_{10}\text{F}_4\text{O}_3$. Calculated, %: C 51.81; H 3.62; F 27.32. M 278.0561.

1,1,2,2,3,3,4,6,7-Nonafluoroindan-5-yl(phenyl)-methanone (XX). A mixture of 7.14 g (10.0 mmol) of organozinc compounds **XIX** [a solution in DMF containing 96% of **XIX** and 4% of 1,1,2,2,3,3,4,5,7-nonafluoroindan (**XXXII**)], 2.11 g (15.0 mmol) of benzoyl chloride (**Ia**), and 0.01 g (0.10 mmol) of CuCl was heated for 4 h at 60–65°C under argon. The mixture was cooled to room temperature, 10 ml of water was added, and the mixture was subjected to steam distillation. A solid crystallized in the receiver and was filtered off and dried in air. According to the GLC data, the product, 1.87 g, contained 85% of ketone **XX** (yield 41%). Sublimation of the still residue gave an additional portion, 0.21 g, of **XX**. After recrystallization from hexane, yield 1.15 g (35%), purity ~99% (GLC), mp 82.5–83.5°C. IR spectrum, ν , cm^{-1} : 3070, 1690, 1599, 1583, 1495, 1452, 1339, 1231, 946. UV spectrum, λ_{max} , nm ($\log \epsilon$): 203 (4.41), 261 (4.11). ^1H NMR spectrum, δ , ppm: 7.53 m (2H, 3'-H, 5'-H), 7.68 m (1H, 4'-H), 7.82 m (2H, 2'-H, 6'-H). ^{19}F NMR spectrum, δ_{F} , ppm: 21.8 d.d.t (1F, 7-F), 31.7 quintet (2F, 2-F), 38.8 d.d (1F, 6-F), 45.9 d.d.t (1F, 4-F), 53.9 m and 54.7 m (2F, 1-F, 3-F); J_{FF} , Hz: 6-F–7-F 21.6, 4-F–7-F 21.1, 3-F–4-F 7.0, 1-F–7-F 7.0, 4-F–6-F 5.2, 2-F–1(3)-F 4.3. Mass spectrum, m/z (I_{rel} , %): 385 (5), 384 (28) $[M]^+$, 307 (9) $[M - \text{Ph}]^+$, 279 (2) $[M - \text{COPh}]^+$, 105 (100) $[\text{COPh}]^+$, 77 (43) $[\text{Ph}]^+$, 51 (13). Found, %: C 50.25; H 1.36; F 44.67. M^+ 384.0190. $\text{C}_{16}\text{H}_5\text{F}_9\text{O}$. Calculated, %: C 50.02; H 1.31; F 44.50. M 384.0191.

1-(1,1,2,2,3,3,4,6,7-Nonafluoroindan-5-yl)propan-1-one (XXI). A mixture of 7.14 g (10.0 mmol) of organozinc compounds **XIX** [a solution in DMF containing 96% of **XIX** and 4% of 1,1,2,2,3,3,4,5,7-nonafluoroindan (**XXXII**)], 1.39 g (15.0 mmol) of propionyl chloride **Ic**, and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 60–65°C under argon. The mixture was cooled to room temperature, treated with 10 ml of

water, and subjected to steam distillation. We isolated 1.94 g of a material containing (GLC) 87% of ketone **XXI** (yield 50%). The product was distilled under reduced pressure (~3 mm), and a fraction with bp 68–70°C (1.01 g) was collected; it contained (GLC) ~98% of ketone **XXI** (yield 30%). IR spectrum, ν , cm^{-1} : 2987, 2944, 1727, 1650, 1492, 1352, 1351, 1257, 1205, 1163, 1126, 1048. UV spectrum, λ_{max} , nm ($\log \epsilon$): 207 (3.97), 276 (3.24). ^1H NMR spectrum, δ , ppm: 1.25 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 2.86 q (2H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz). ^{19}F NMR spectrum, δ_{F} , ppm: 21.9 d.d.t (1F, 7-F), 31.6 quint (2F, 2-F), 37.0 d.d (1F, 6-F), 44.2 d.d.t (1F, 4-F), 53.7 m and 54.5 m (2F, 1-F, 3-F); J_{FF} , Hz: 6-F–7-F 21.4, 4-F–7-F 21.4, 3-F–4-F 6.9, 1-F–7-F 6.9, 4-F–6-F 4.4, 2-F–1(3)-F 4.0 Hz. Mass spectrum, m/z (I_{rel} , %): 337 (1), 336 (9) $[M]^+$, 307 (100) $[M - \text{Et}]^+$, 279 (2) $[M - \text{COEt}]^+$, 57 (3) $[\text{COEt}]^+$, 29 (4) $[\text{Et}]^+$. Found, %: C 43.42; H 1.46; F 50.86. M^+ 336.0195. $\text{C}_{12}\text{H}_5\text{F}_9\text{O}$. Calculated, %: C 42.88; H 1.50; F 50.87. M 336.0191.

Phenyl(2,3,5,6-tetrafluoropyridin-4-yl)methanone (XXIII). A mixture of 8.85 g (10.0 mmol) of organozinc compounds **XXII** (a solution in DMF preliminarily treated with CaCl_2 ; fraction of **XXII** 93%, fraction of **XXV** 7%), 1.55 g (11.0 mmol) of benzoyl chloride (**Ia**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 55–60°C under argon. The mixture was cooled to room temperature, 10 ml of water was added, and the mixture was subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.86 g of a mixture containing (92%) of ketone **XXIII** (yield 76%) and 2% of **XXV**. Vacuum sublimation at 75°C (~2 mm) gave 1.65 g (65%) of ketone **XXIII** with a purity of ~98% (GLC). IR spectrum, ν , cm^{-1} : 3067, 2610, 1694, 1599, 1583, 1492, 1465, 1411, 1289, 970, 686, 683. UV spectrum, λ_{max} , nm ($\log \epsilon$): 204 (4.09), 261 (4.11). ^1H NMR spectrum, δ , ppm: 7.52 m (2H, 3-H, 5-H), 7.67 m (1H, 4-H), 7.81 m (2H, 2-H, 6-H) [11]. ^{19}F NMR spectrum, δ_{F} , ppm: 19.6 m (2F, 3'-F, 5'-F), 73.6 m (2F, 2'-F, 6'-F). Mass spectrum, m/z (I_{rel} , %): 256 (5), 255 (38) $[M]^+$, 178 (2) $[M - \text{Ph}]^+$, 150 (4) $[M - \text{COPh}]^+$, 105 (100) $[\text{COPh}]^+$, 77 (73) $[\text{Ph}]^+$, 51 (30). Found, %: C 56.03; H 1.99; F 30.48; N 5.18. M^+ 255.0304. $\text{C}_{12}\text{H}_5\text{F}_4\text{NO}$. Calculated, %: C 56.48; H 1.98; F 29.78; N 5.49. M 255.0307.

1-(2,3,5,6-Tetrafluoropyridin-4-yl)propan-1-one (XXIV). *a.* A mixture of 9.05 g (14.48 mmol) of organozinc compounds **XXII** (97% of **XXII** and 3% of **XXV**; a solution in DMF without preliminary treatment with CaCl_2), 2.78 g (30.0 mmol) of propionyl

chloride (**1c**), and 0.015 g (0.15 mmol) of CuCl was heated for 4 h at 60–65°C under argon. According to the ^{19}F NMR data, the mixture contained ~28% of ketone **XXIV**, ~18% of **XXV**, ~5% of **XXVI**, ~2% of **XXVII**, and 47% of propionyl fluoride (**1f**; ^{19}F NMR spectrum: δ 205.7 ppm, s [20]). The mixture was cooled to room temperature, treated with 10 ml of water, and steam-distilled. The organic phase was separated and dried over MgSO_4 . The product, 1.71 g, contained (GLC) 47% of ketone **XXIV** (yield 27%), 7% of compound **XXV**, and 6% of **XXVI**.

b. Likewise, the reaction mixture obtained from 9.38 g (14.73 mmol) of organozinc compounds **XXII** (98% of **XXII** and 2% of **XXV**; a solution in DMF without preliminary treatment with CaCl_2) and 4.16 g (44.9 mmol) of propionyl chloride (**1c**) in the presence of 0.015 g (0.15 mmol) of CuCl contained (according to the ^{19}F NMR data) ~41% of ketone **XXIV**, ~12% of **XXV**, ~1% of **XXVI**, ~2% of **XXVII**, and 44% of propionyl fluoride (**1f**). After appropriate treatment, we isolated 2.28 g of a mixture consisting (GLC) of 82% of ketone **XXIV** (yield 62%), 5% of **XXV**, and 2% of **XXVI**. The mixture was distilled under reduced pressure (~2 mm), and a fraction with bp 63°C (0.88 g) was collected; it contained (GLC) ~99% of ketone **XXIV** (yield 29%). IR spectrum, ν , cm^{-1} : 2982, 2946, 2888, 1842, 1730, 1642, 1465, 1409, 1269, 956, 859. UV spectrum: λ_{max} 275 nm ($\log \epsilon$ 3.46). ^1H NMR spectrum, δ , ppm: 1.25 t (3H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz), 2.86 q (2H, CH_2CH_3 , $J_{\text{HH}} = 7.2$ Hz). ^{19}F NMR spectrum, δ , ppm: 17.9 m (2F, 3-F, 5-F), 73.6 m (2F, 2-F, 6-F). Mass spectrum, m/z (I_{rel} , %): 208 (6), 207 (59) $[M]^+$, 178 (99) $[M - \text{Et}]^+$, 150 (41) $[M - \text{COEt}]^+$, 100 (32), 57 (100) $[\text{COEt}]^+$, 29 (82) $[\text{Et}]^+$. Found, %: C 46.35; H 2.36; F 36.54; N 6.95. M^+ 207.0313. $\text{C}_8\text{H}_5\text{F}_4\text{NO}$. Calculated, %: C 46.39; H 2.43; F 36.69; N 6.76. M 207.0307.

c. A mixture of 8.85 g (10.0 mmol) of organozinc compounds **XXII** (a solution in DMF preliminarily treated with CaCl_2 ; fraction of **XXII** 93%, fraction of **XXV** 7%), 1.02 g (11.0 mmol) of propionyl chloride (**1c**), and 0.01 g (0.10 mmol) of CuCl was heated for 2 h at 55–60°C under argon. The mixture was cooled to room temperature, treated with 10 ml of water, and subjected to steam distillation. The organic phase was separated and dried over MgSO_4 . We isolated 1.70 g of a mixture containing (GLC) 83% of ketone **XXIV** (yield 68%) and 14% of compound **XXV**.

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